From Sequences to Biological Insight - a protein perspective

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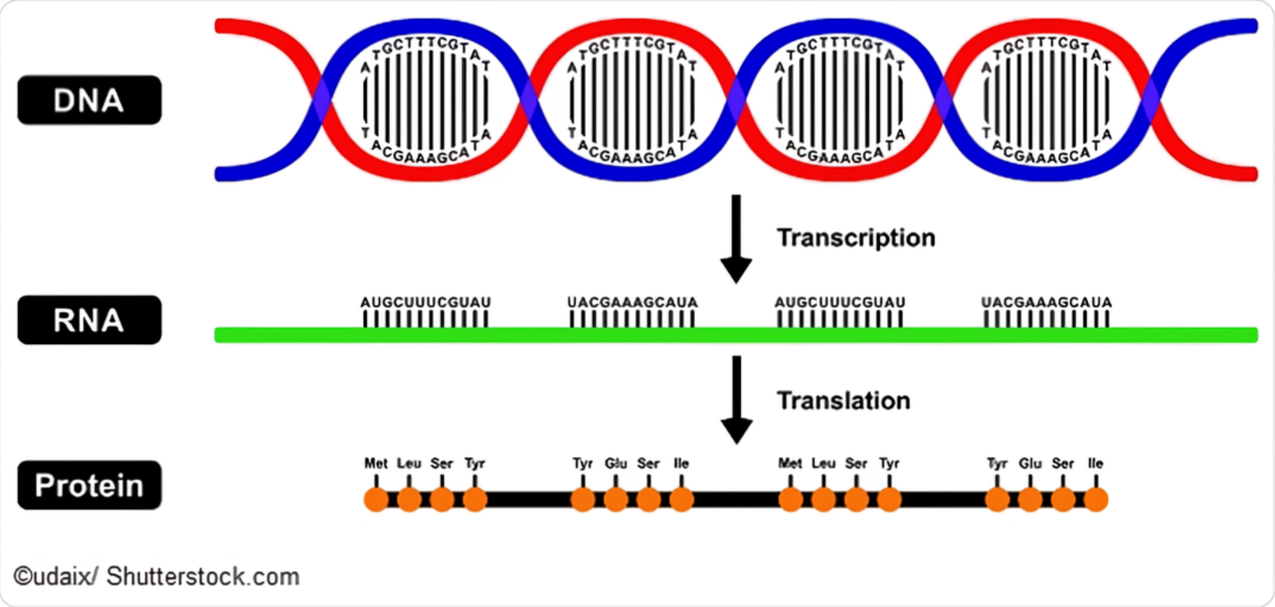
Outline of today

- DNA/protein sequences
- Sequence alignment
- Sequence conservation and evolution

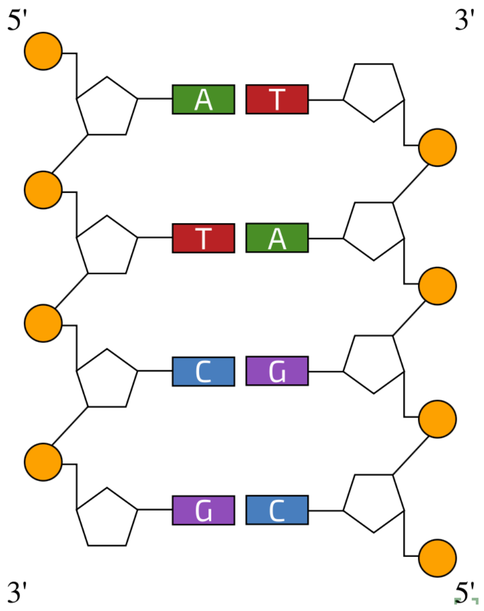
Why do we analyze biological sequences?

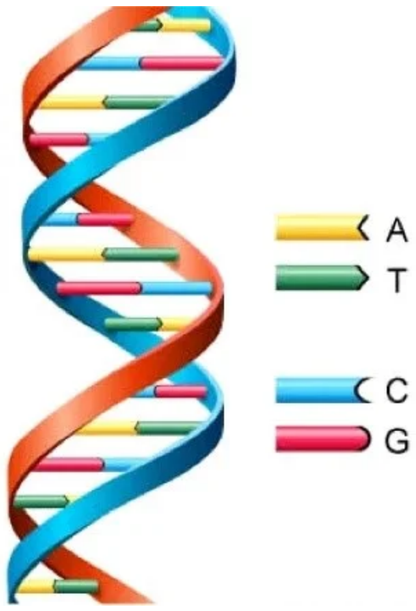
- Inherited genetic material = genome
- encodes mRNA and proteins
- Genome
  - defines the difference between species
  - affects disease susceptibility



DNA (deoxyribonucleic acid) sequence

- Carrier of genomic information for most species
- Building block: nucleotides (bases)
  - Adenine (A)
  - Thymine (T)
  - Cytosine (C)
  - Guanine (G)

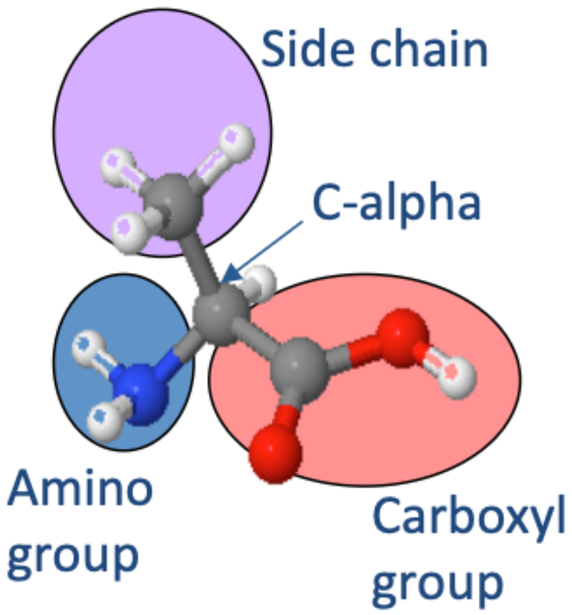


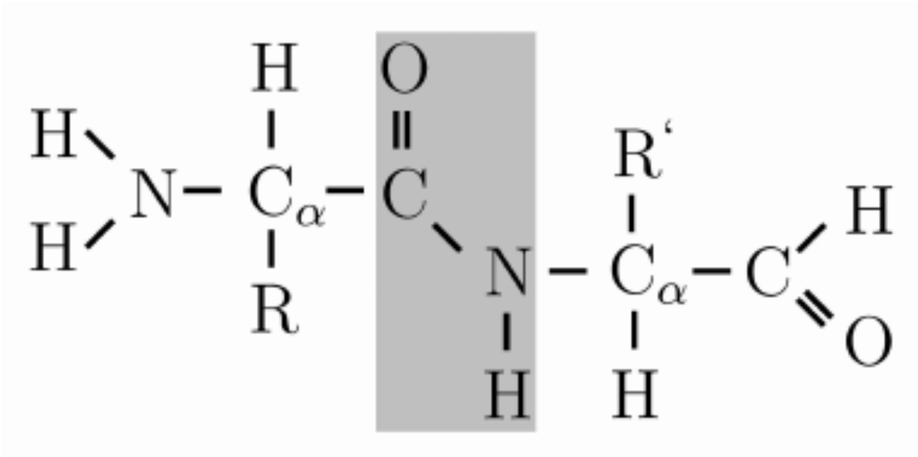


- A coding sequence typically 900-1300 bp, but spans 10-100 kbp in eukaryotes due to introns

Protein

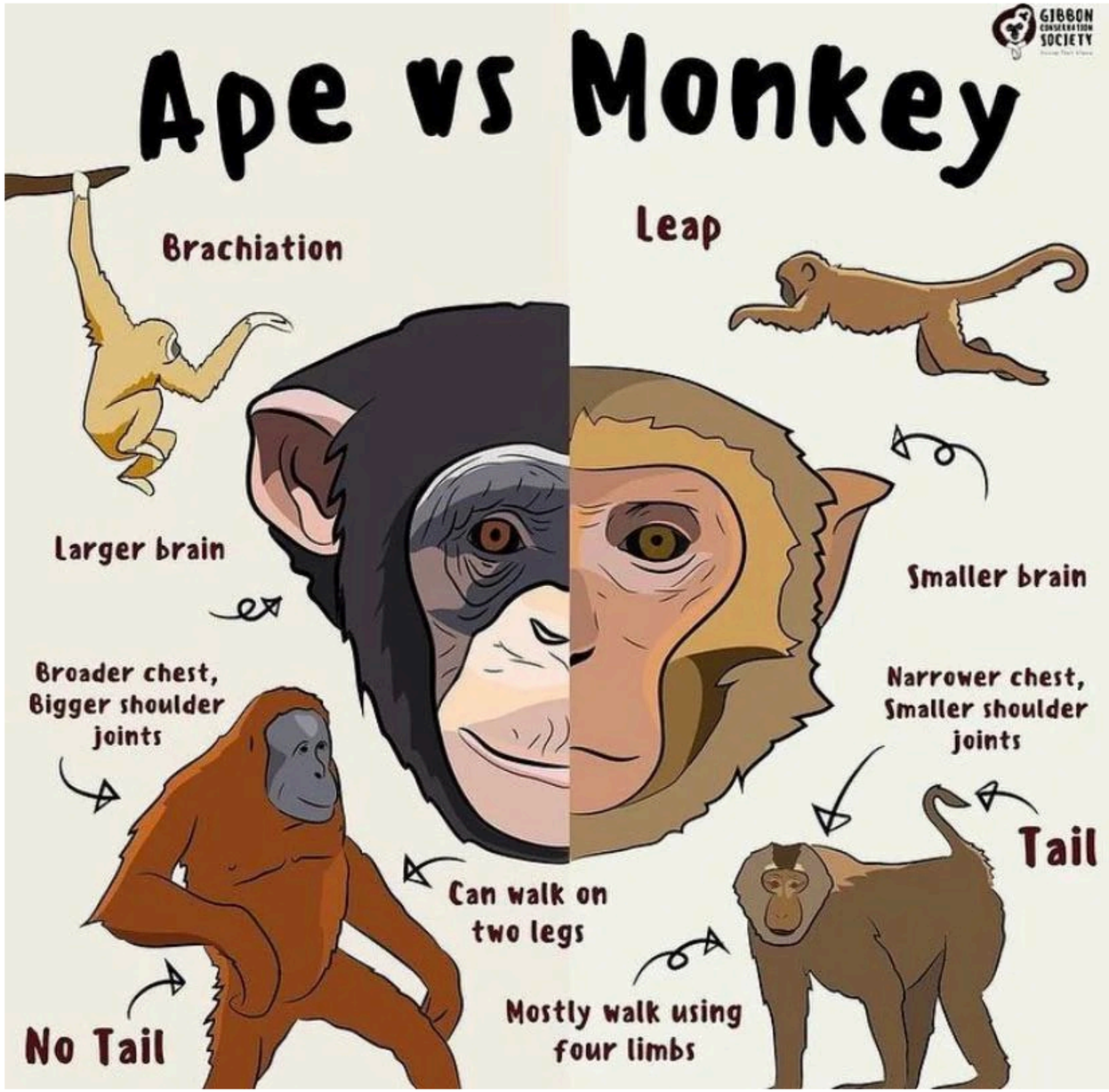
- translation product
- Building block = amino acids (AAs)
  - 20 AAs
  - More combinations / information
    - 2 nucleotides = 4x4 =16
    - 2 amino acides = 20 x 20 = 400
- Protein length typically 300-500 AAs





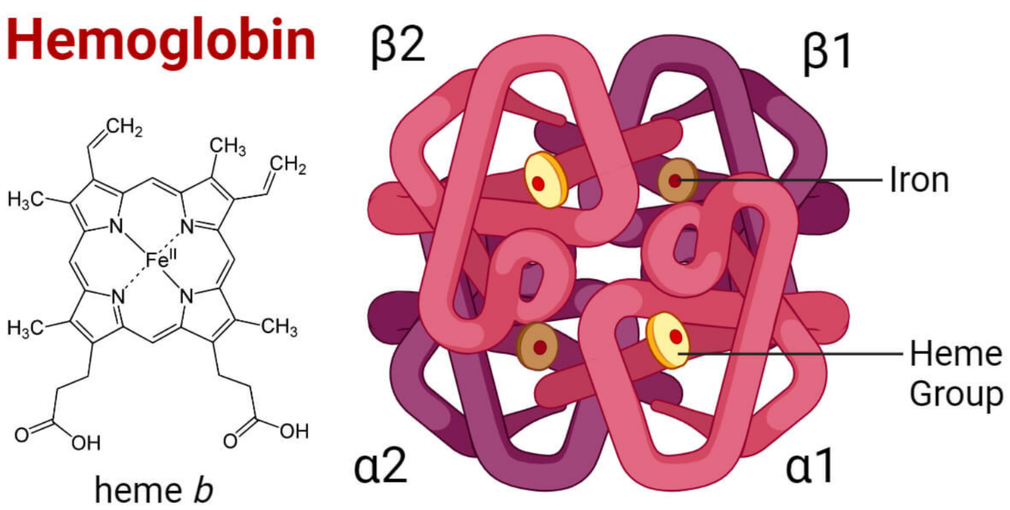
Genome defines the difference between species/individuals

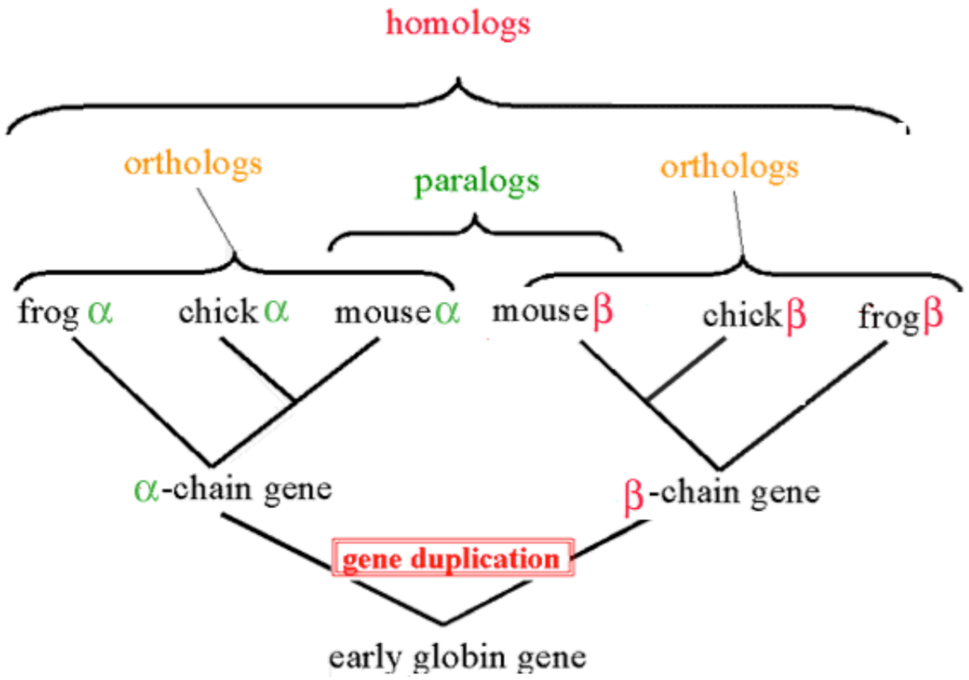
- How much genetic material does a macaque share with a chimp?
- How do the genes responsible for smelling differ between a human and a dog?
- How/where does a disease-causing gene differ from a healthy gene?

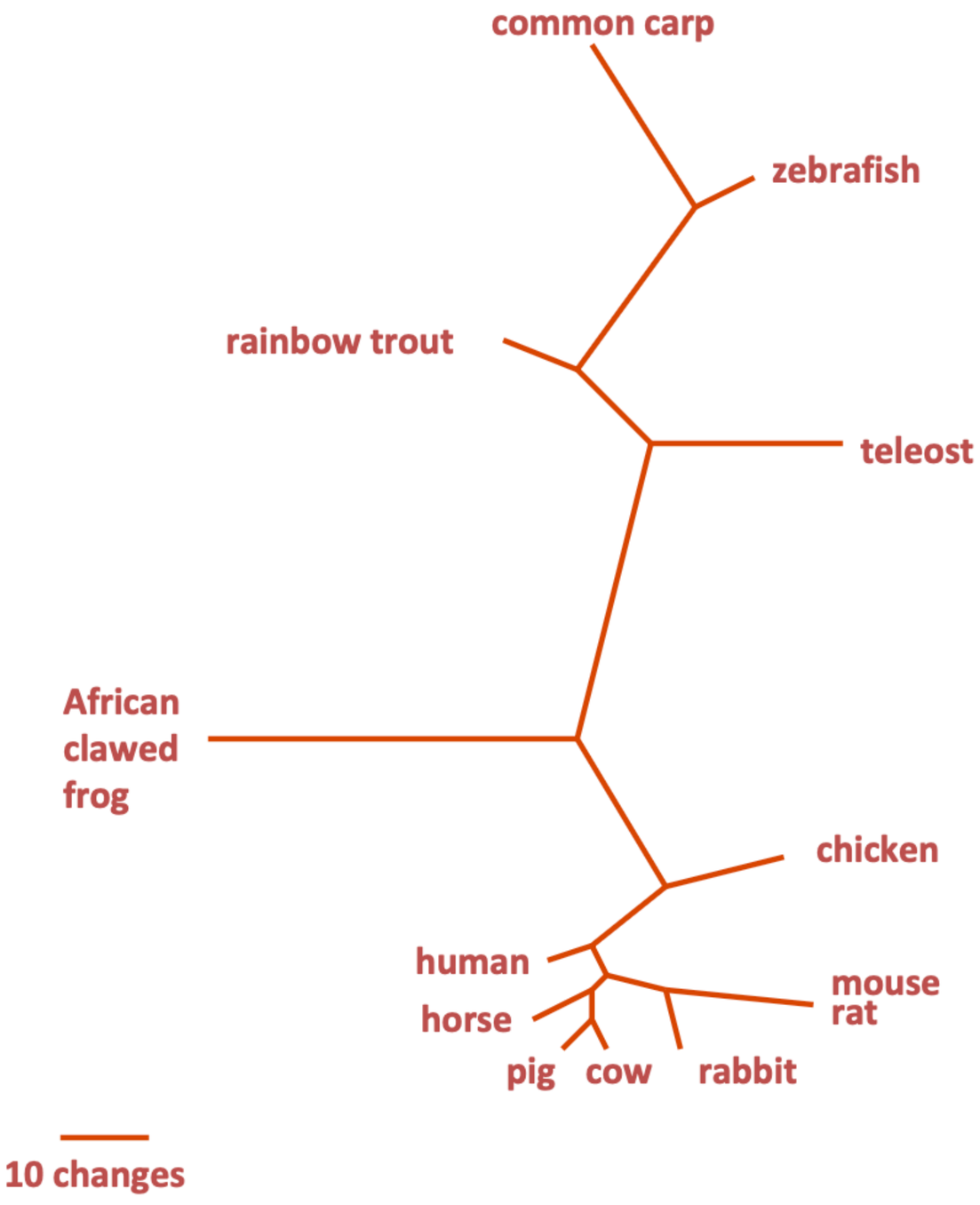


Homology

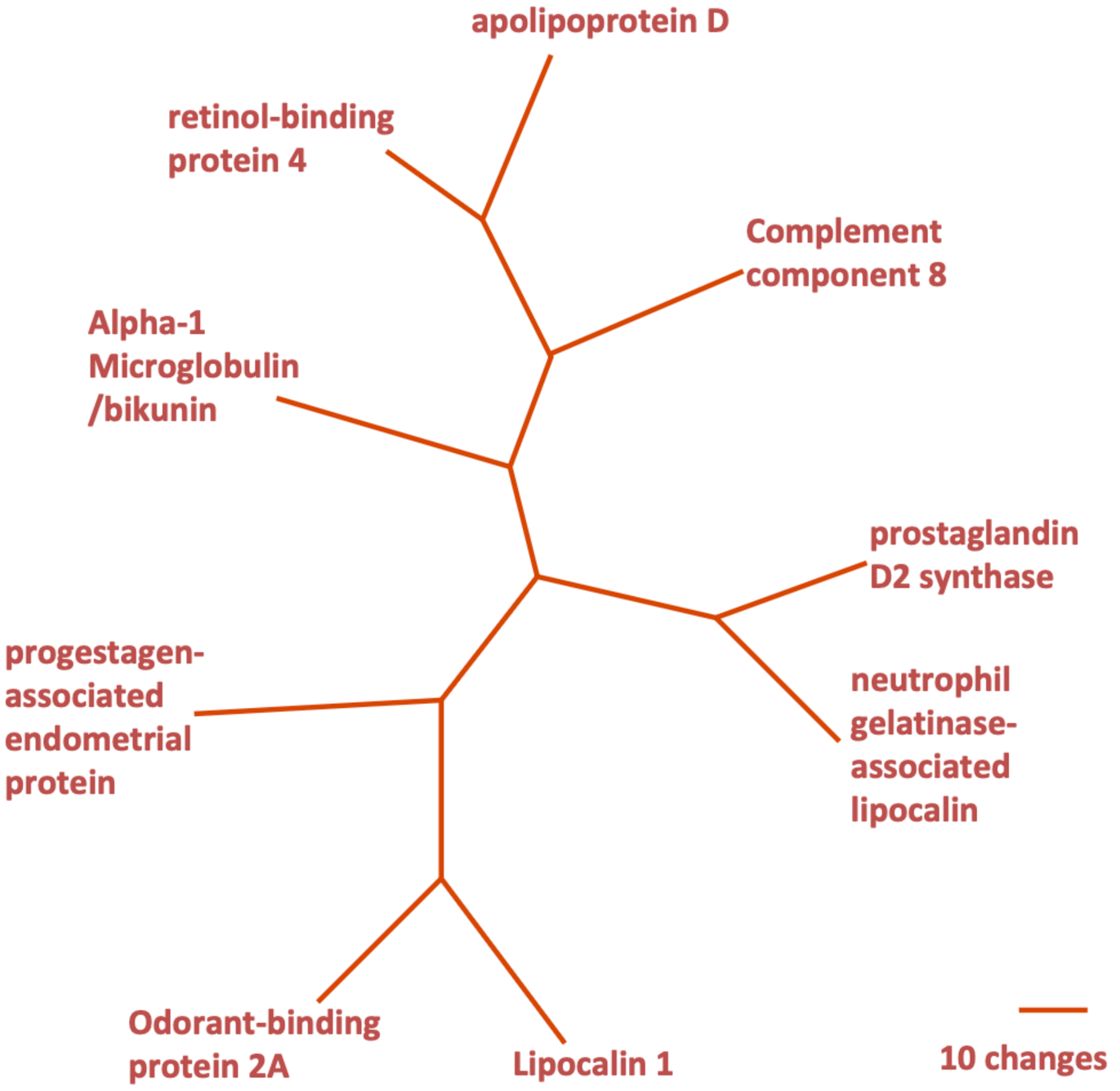
- Similarity attributed to descent from a common ancestor.
- Homolog
  - Ortholog (divergence by **speciation**)
  - Paralog (divergence by **gene duplication**)







Orthologs: members of a gene (protein) family in various organisms. This tree shows RBP orthologs.



Paralogs: members of a gene (protein) family within a species

Sequence alignment

- Arrange biological sequences in rows to compare sequences and identify regions of similarity
  - DNA sequence
  - Protein sequence

DNA (nucleotide) alignment

- Better suited
  - analyze mutations/SNPs
  - confirm the identity of a cDNA
  - study non-coding regions of DNA
  - compare closely related species/individuals

```
Query: 181 catcaactacaactccaaagacacccttacacccactaggatatcaacaaacctacccac 240
           |||||||| |||| |||||| ||||| | ||||||||||||||||||||||||||||||
Sbjct: 189 catcaactgcaaccccaaagccacccct-cacccactaggatatcaacaaacctacccac 247
```

Neanderthal vs modern human DNA

Protein sequence alignment

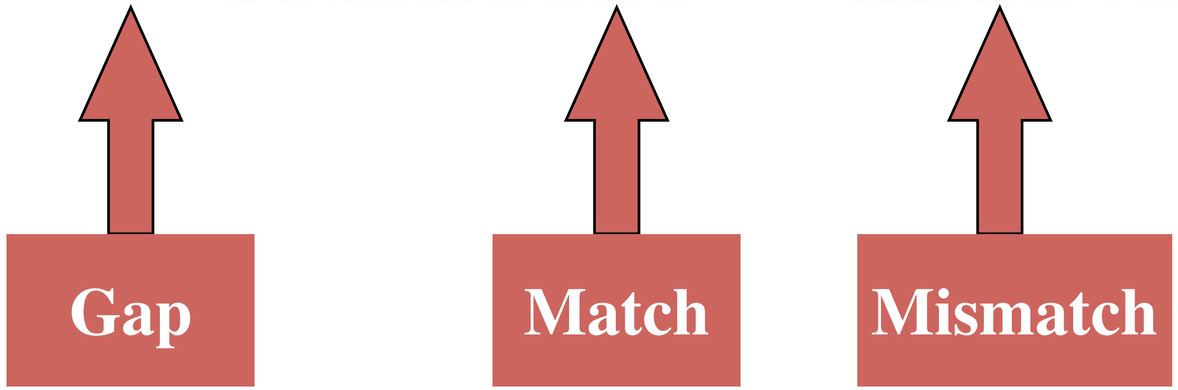
- Protein sequence is more conserved/robust on functional level (codon redundancy)

  e.g. GCU, GCC, GCA, GCG → Alanine

- Better suited:
  - Compare distant homologs
  - Detect conserved domains/motifs
- Technically different from DNA: varying degrees of similarity between amino acids
  - # DNA mutations to convert
  - biochemical properties
  - mutation frequencies

Example DNA sequence alignment

```
ACGTCTGATACGCCGTATAGTCTATCT
    ||||| |||   || ||||||||
----CTGATTCGC---ATCGTCTATCT
```



Positions at which a letter is paired with a null

Example protein sequence alignment

Match

```
  1 MKWVWALLLLAAWAAAERDCRVSSFRVKENFDKARFSGTWYAMAKKDPEG 50 RBP
      .  |||  |     .   |.  .  .  |  : .||||.:|    :
  1 ...MKCLLLALALTCGAQALIVT..QTMKGLDIQKVAGTWYSLAMAASD. 44 lactoglobulin

 51 LFLQDNIVAEFSVDETGQMSATAKGRVR.LLNNWD.  ADMVGTFTDTE 97 RBP
    : | |   |   |    ::  | .| .  ||  |:  ||        |.
 45 ISLLDAQSAPLRV.YVEELKPTPEGDLEILLQKWENC CAQKKIIAEKTK 93 lactoglobulin

 98 DPAKFKMKYWGVASFLQKGNDDHWIVDTDYDTYAV. ........QYSC 136 RBP
     || ||.        |        :.||||  | .            .|
 94 IPAVFKIDALNENKVL........VLDTDYKKYLLFC ENSAEPEQSLAC 135 lactoglobulin

137 RLLNLDGTCADSYSFVFSRDPNGLPPEAQKIV          RQYRLIV 185 RBP
    . |       |     | :    ||   .
136 QCLVRTPEVDDEALEKFDKALKALPMHIRLSFNPTQLEEQCHI....... 178 lactoglobulin
```

Identity (bar)

Example protein sequence alignment

Mismatch

```
  1 MKWVWALLLLAAWAAAERDCRVSSFRVKENFDKARFSGTWYAMAKKDPEG 50 RBP
       .  |||  |     .   |.  .  .  |  : .||||.:|    :
  1 ...MKCLLLALALTCGAQALIVT..QTMKGLDIQKVAGTWYSLAMAASD. 44 lactoglobulin

 51 LFLQDNIVAEFSVDET MSATAKGRVR.LLNNWD..VCADMVGTFT E 97 RBP
    : | |   |   |    :  | .| .  ||  |:   ||        .
 45 ISLLDAQSAPLRV.YVK LKPTPEGDLEILLQKWENGECAQKKIIAE K 93 lactoglobulin

 98 DPAKFKMKYWGVASFLQ GNDDHWIVDTDYDTYAV..........Q C 136 RBP
     || ||.        |       :.||||  | .            |
 94 IPAVFKIDALNENKVL .....VLDTDYKKYLLFCMENSAEPEQS C 135 lactoglobulin

137 RLLNLDGTCA PPEAQKIVRQRQ.EELC RBP
    . |        |   .      | || |
136 QCLVRTPEVD PMHIRLSFNPTQLEEQC lactoglobulin
```

Somewhat similar (one dot)

Very similar (two dots)

Example protein sequence alignment

Gaps

```
  1 MKWVWALLLLAAWAAAERDCRVSSFRVKENFDKARFSGTWYAMAKKDPEG 50 RBP
       .  |||  |     .   |.  .  .  |  : .||||.:|    :
  1 ...MKCLLLALALTCGAQALIVT..QTMKGLDIQKVAGTWYSLAMAASD. 44 lactoglobulin

 51 LFLQDNIVAEFSVDETGQMSATAKGRVR.LLNNWD..VCADMVGTFTDTE 97 RBP
    : | |   |   |    ::  | .| .  ||  |:   ||        |.
 45 ISLLDAQSAPLRV.YVEELKPTPEGDLEILLQKWENGECAQKKIIAEKTK 93 lactoglobulin

 98 DPAKFKMKYWGVASFLQKGNDDHWIVDTDYDTYAV...........QYSC 136 RBP
     || ||.        |        :.||||  | .             .|
 94 IPAVFKIDALNENKVL........VLDTDYKKYLLFCMENSAEPEQSLAC 135 lactoglobulin

137 RLLNLDGTCADSYSFVFSL  NGLPPEAQKIVRQRQ.EELCLARQYRLIV 185 RBP
    . |       |     | :    ||   .      | || |
136 QCLVRTPEVDDEALEKFDK  KALPMHIRLSFNPTQLEEQCHI....... 178 lactoglobulin
```

Internal gap

Terminal gap

Sequence comparison relies on alignment

- Locate equivalent regions between sequences
- Maximize the number of matches (find the optimal superposition)
- Finally, derive
  - **Identity** (percentage match)
  - **Conservation** (percentage functional match, in case of AA)
  - **Similarity** (combining the two)

```
THISSEQUENCE
THISISASEQUENCE
```

❌

```
THIS-SEQUENCE
THISISASEQUENCE
```

❌

```
THIS--SEQUENCE
THISISASEQUENCE
```

❌

```
THIS---SEQUENCE
THISISASEQUENCE
```

✅

How to find an optimal alignment?

Score the Alignments

- Quantify the goodness of alignments
- A simple example with Match +1, Mismatch 0
- Compute the sum on all positions

	A	T	C	G
A	1	0	0	0
T	0	1	0	0
C	0	0	1	0
G	0	0	0	1

```
AATCTATA
AAGATA
```
4

```
AATCTATA
AAGATA
```
1

```
AATCTATA
  AAGATA
```
3

```
THISSEQUENCE
THISISASEQUENCE
```
4

```
THIS-SEQUENCE
THISISASEQUENCE
```
5

```
THIS--SEQUENCE
THISISASEQUENCE
```
4

```
THIS---SEQUENCE
THISISASEQUENCE
```
12

Score the Alignments with gap penalties

- Gap opening is worse than gap extension
- Because we want to keep the matches in blocks
- A simple example Gap penalty = -5, Gap extension penalty = -2

```
AATCTATA
AAGA--TA
```

−3

```
AATCTATA
AA--GATA
```

−2

```
AATCTATA
AA-G-ATA
```

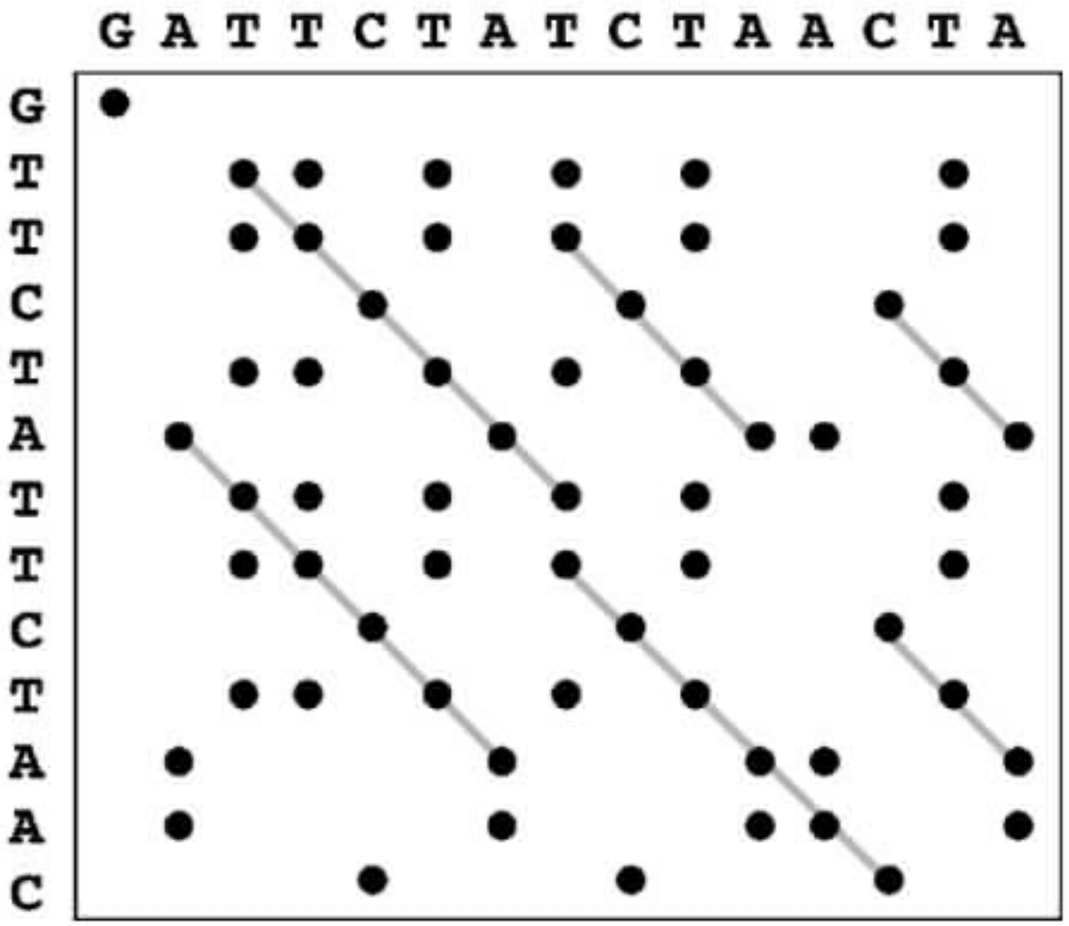
−5

How to find an optimal alignment

- Finding the alignment is computationally hard
  ```
  ACGTCTGATACGCCGTATAGTCTATCT
  CTGAT---TCG-CATCGTC--T-ATCT
  ```
  - C(27,7) gap positions = ~888,000 possibilities
- The first formal algorithm: Needleman & Wunsch algorithm (1970) using dynamic programming

Dot plot method (1970s)

- One sequence is placed on the x-axis, the other on the y-axis
- Dots:matches
- Diagonal lines: consecutive matches
- Visualizes
  - Diagonal: similarity
  - Parallel: repeats



Example of comparing two sequences using dot plots. (Xiong, J., 2006).

Needleman-Wunsch algorithm

Dynamic programming (DP)

- Define the score: Match +1, Mismatch 0, Gap -1

	A	T	C	G
A	1	0	0	0
T	0	1	0	0
C	0	0	1	0
G	0	0	0	1

		A	A	T	C	T	A	T
	0	−1	−2	−3	−4	−5	−6	−7
A	−1							
A	−2							
G	−3							
A	−4							
T	−5							
A	−6							

Needleman-Wunsch algorithm

- Vertical/Horiz. move: Score + (simple) gap penalty
- Diagonal move: Score + match/mismatch score
- Take the MAX

		A	A	T	C	T	A	T
	0	−1	−2	−3	−4	−5	−6	−7
A	−1							
A	−2							
G	−3							
A	−4							
T	−5							
A	−6							

Needleman-Wunsch algorithm

- Vertical/Horiz. move: Score + (simple) gap penalty
- Diagonal move: Score + match/mismatch score
- Take the MAX

Vertical: -1 - 1 = -2
Horizontal: -1 - 1 = -2
Diagonal: 0 + 1 = 1

		A	A	T	C	T	A	T
	0	−1	−2	−3	−4	−5	−6	−7
A	−1	1						
A	−2							
G	−3							
A	−4							
T	−5							
A	−6							

Needleman-Wunsch algorithm

Backtrace

- Once the table is filled
  - go from the lower right corner
  - Return to 0 position with arrows

		g	c	g
	0	-1	-2	-3
g	-1	1	0	-1
g	-2	0	1	1
c	-3	-1	1	1
g	-4	-2	0	2

Equally valid alignments with score 2

```
ggcg        ggcg
g-cg        -gcg
```

Exercise 1

- Fill the table
- Find the optimal alignment

		g	c	g
	0	-1	-2	-3
g	-1	1	0	-1
g	-2	0	1	1
c	-3	-1	1	1
g	-4	-2	0	2

		A	A	T	C	T	A	T
	0	−1	−2	−3	−4	−5	−6	−7
A	−1	1						
A	−2							
G	−3							
A	−4							
T	−5							
A	−6							

DNA scoring parameters

- Typically not the simple identity matrix
  - Match: +1
  - Mismatch: -2 (or -3)
  - Gap open: -5
  - Gap extension: -2

	A	T	C	G
A	1	−2	−2	−2
T	−2	1	−2	−2
C	−2	−2	1	−2
G	−2	−2	−2	1

Protein scoring - substitution matrix

PAM (Point Accepted Mutation)

- A pair of amino acids
  - The more chemically similar
  - The higher the **exchange odds**
- PAM1
  - 1,572 observed mutations across 71 families of closely related proteins (1% divergence)
- PAM250
  - extrapolated from PAM1 by 250 times
  - to predict the odds on very distant sequences

PAM250 matrix

	C	S	T	P	A	G	N	D	E	Q	H	R	K	M	I	L	V	F	Y	W
C	12																			
S	0	2																		
T	-2	1	3																	
P	-3	1	0	6																
A	-2	1	1	1	2															
G	-3	1	0	-1	1	5														
N	-4	1	0	-1	0	0	2													
D	-5	0	0	-1	0	1	2	4												
E	-5	0	0	-1	0	0	1	3	4											
Q	-5	-1	-1	0	0	-1	1	2	2	4										
H	-3	-1	-1	0	-1	-2	2	1	1	3	6									
R	-4	0	-1	0	-2	-3	0	-1	-1	1	2	6								
K	-5	0	0	-1	-1	-2	1	0	0	1	0	3	5							
M	-5	-2	-1	-2	-1	-3	-2	-3	-2	-1	-2	0	0	6						
I	-2	-1	0	-2	-1	-3	-2	-2	-2	-2	-2	-2	-2	2	5					
L	-6	-3	-2	-3	-2	-4	-3	-4	-3	-2	-2	-3	-3	4	2	6				
V	-2	-1	0	-1	0	-1	-2	-2	-2	-2	-2	-2	-2	2	4	2	4			
F	-4	-3	-3	-5	-4	-5	-4	-6	-5	-5	-2	-4	-5	0	1	2	-1	9		
Y	0	-3	-3	-5	-3	-5	-2	-4	-4	-4	0	-2	-4	-2	-1	-1	-2	7	10	
W	8	-2	-5	-6	-6	-7	-4	-7	-7	-5	-3	-2	-3	-4	-5	-2	-6	0	0	17

BLAST

- Needleman-Wunsch is too slow for the database searches **O(n·m)** time
- “Find similar sequences in a huge database quickly, not perfectly optimally.”
- **BLAST: Heuristic local** alignment search tool (1990)
- much faster

BLAST

seed-and-extend

- Step 1: Break query into “words”
  - Example (protein BLAST default word size = 3 amino acids):
  - Query: MKTAYIAKQR
  - Words: MKT, KTA, TAY, ...
- Step 2: Find exact or near-exact word matches in database
  - Quickly scans database for matches to these “seeds”.
- Step 3: Extend matches
  - If a word match is found, BLAST extends alignment in both directions
  - Uses a scoring system (like substitution matrices)
- Step 4: Score and filter
  - Keeps only high-scoring segment pairs (HSPs)
  - Reports statistically significant hits (E-values)

Global VS local alignment

Use cases

- Global - “Do these two full genes/proteins correspond overall?”
  - Reference gene vs patient gene
  - assembled contig vs reference genome
- Local - “Do these sequences share any meaningful region?”
  - Find conserved domains or motifs
  - Search database for homologs

Global VS local alignment

Algorithms

- Global - “Do these two full genes/proteins correspond overall?”
  - Needleman-Wunsch
  - EMBOSS-Stretcher
- Local - “Do these sequences share any meaningful region?”
  - BLAST
  - DIAMOND (2015) - 20,000x faster than BLAST

Exercise 2

- Run EMBOSS-Stretcher for these two sequences online
  - THISISASEQENCE and THISSEQENCE
  - What's the default parameters used? **Matrix, gap penalty, extend penalty**
  - Derive the computed score as in the output
- Run BLAST for a human protein online
  - Go to Uniprot database and find AXIN1_HUMAN entry
  - Paste its sequence to BLASTp
  - What are the top 10 proteins?
  - Use your own words to explain **Expect (E-value), bit score, %identity, alignment length**

BLAST output interpretation

- Expect (E-value): The number of matches you'd expect to see by chance in a database of that size.
  - Lower = better
  - Larger database = higher E-value
- Bit score: A normalized score representing the **quality of the alignment** (comparable across different searches)
  - Independent of database size, only alignment score between the query/reference
  - Higher = better
- %Identity: The percentage of positions in the alignment that are **exactly the same**.
- Alignment length: The number of positions included in the alignment (including matches, mismatches, and gaps)
  - Longer = more meaningful

BLOSSUM (blocks substitution matrix)

- Derived from local, ungapped alignments of distantly related sequences
- BLOSUM62 (default) calculated from sequences clustered with no more than 62% identity.
- BLOSUM80 for closer sequences
- BLOSUM45 for distant sequences
- Henikoff and colleagues (PNAS 89:10915)
- More often used than PAM as it's directly derived from data

Homework

- We talked about hemoglobin, consisting of 4 subunits.
- In human, there are three genes HBA1, HBA2 and HBB genes
- Find out how similar are these genes on **both DNA and protein level** using appropriate local alignment tool
- And do the same for mouse (*mus musculus*)

Resources

https://www.slideserve.com/helen/sequence-alignments-powerpoint-ppt-presentation

PAM and BLOSUM scoring matrices: https://www.youtube.com/watch?v=MVFVGKebN9U

Hemoglobin https://microbenotes.com/hemoglobin/

Evolution of Hemoglobin and Its Genes https://pmc.ncbi.nlm.nih.gov/articles/PMC3543078/